Unit: 6 Unit 2: Exponents Radicals
Topic: Rationalizing Denominator

Name: ________________
Date: ____________

Advanced (Habanero)

Q1. Rationalize the denominator: $\frac{1}{\sqrt{2}}$

(A) $\frac{\sqrt{2}}{2}$
(B) $\frac{1}{2\sqrt{2}}$
(C) $\frac{2}{\sqrt{2}}$
(D) $\frac{\sqrt{2}}{\sqrt{2}}$
(E) $\frac{1}{2}$

Q2. Simplify: $\frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$

(A) $\frac{1}{3}$
(B) $\frac{\sqrt{3}}{3}$
(C) $\frac{3}{\sqrt{3}}$
(D) $\frac{\sqrt{3}}{\sqrt{3}}$
(E) $\frac{1}{\sqrt{3}}$

Q3. Rationalize the denominator: $\frac{2}{\sqrt{5}-1}$

(A) $\frac{2(\sqrt{5}+1)}{4}$
(B) $\frac{2(\sqrt{5}-1)}{4}$
(C) $\frac{2(\sqrt{5}+1)}{5-1}$
(D) $\frac{2(\sqrt{5}+1)}{5}$
(E) $\frac{2(\sqrt{5}-1)}{5}$

Q4. Rationalize the denominator: $\frac{3}{\sqrt{7}+2}$

(A) $\frac{3(\sqrt{7}-2)}{3}$
(B) $\frac{3(\sqrt{7}+2)}{3}$
(C) $\frac{3(\sqrt{7}-2)}{7-4}$
(D) $\frac{3(\sqrt{7}+2)}{7}$
(E) $\frac{3(\sqrt{7}-2)}{7}$

Q5. Simplify: $(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}})^2$

(A) $\frac{5}{6}$
(B) $\frac{2}{3}$
(C) $\frac{1}{2} + \frac{1}{3} + \frac{2\sqrt{6}}{6}$
(D) $\frac{1}{2} + \frac{1}{3} + \frac{2\sqrt{2}}{6}$
(E) $\frac{1}{2} + \frac{1}{3} + \frac{2\sqrt{3}}{6}$

Q6. Rationalize the denominator: $\frac{\sqrt{2}}{\sqrt{6}+\sqrt{3}}$

(A) $\frac{\sqrt{2}(\sqrt{6}-\sqrt{3})}{6-3}$
(B) $\frac{\sqrt{2}(\sqrt{6}+\sqrt{3})}{6-3}$
(C) $\frac{\sqrt{2}(\sqrt{6}+\sqrt{3})}{6+3}$
(D) $\frac{\sqrt{2}(\sqrt{6}-\sqrt{3})}{6+3}$
(E) $\frac{\sqrt{2}(\sqrt{6}+\sqrt{3})}{3}$

Q7. Rationalize the denominator: $\frac{2}{\sqrt{10}+1}$

(A) $\frac{2(\sqrt{10}-1)}{10-1}$
(B) $\frac{2(\sqrt{10}+1)}{10-1}$
(C) $\frac{2(\sqrt{10}+1)}{10}$
(D) $\frac{2(\sqrt{10}-1)}{10}$
(E) $\frac{2(\sqrt{10}+1)}{9}$

Q8. Simplify: $\frac{\sqrt{2}+\sqrt{3}}{\sqrt{2}-\sqrt{3}}$

(A) $\frac{(\sqrt{2}+\sqrt{3})(\sqrt{2}+\sqrt{3})}{2-3}$
(B) $\frac{(\sqrt{2}+\sqrt{3})(\sqrt{2}-\sqrt{3})}{2-3}$
(C) $\frac{(\sqrt{2}+\sqrt{3})(\sqrt{2}+\sqrt{3})}{2+3}$
(D) $\frac{(\sqrt{2}+\sqrt{3})(\sqrt{2}-\sqrt{3})}{2+3}$
(E) $\frac{(\sqrt{2}-\sqrt{3})(\sqrt{2}+\sqrt{3})}{2+3}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator and simplify the expression $\frac{3\sqrt{2}+2}{\sqrt{5}-1} \cdot \frac{\sqrt{5}+1}{\sqrt{5}+1}$

Q10. Simplify the expression $(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}})(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}})$ and rationalize the denominator if necessary

Chef's Tips

- Tip 1: Check if students have rationalized the denominators correctly
- Tip 2: Make sure students simplify their expressions
- Tip 3: Pay attention to the signs in the rationalization process